1D LCM, HCF, and Prime Factorisation

Standard Worksheet

Name: _____ Class: _____ Date: _____

PART A · FLUENCY & SKILL PRACTICE

Show complete step-by-step working in your workbook

Q1. Write the prime factorisation of each number in index form:

(a) 90 (b) 144 (c) 252 (d) 540

Q2. Find the Highest Common Factor (HCF) of each pair or group of numbers:

(a) 48 and 80 (b) 63 and 105 (c) 72 and 120 (d) 36, 60, and 84

Q3. Find the Lowest Common Multiple (LCM) of each pair or group of numbers:

(a) 12 and 15 (b) 18 and 24 (c) 16 and 28 (d) 8, 10, and 12

Q4. Given $A = 2^3 \times 3^2 \times 5$ (360) and $B = 2^2 \times 3^3 \times 7$ (756), find:

(a) $\mathrm{HCF}(A, B)$ in index form (b) $\mathrm{HCF}(A, B)$ as a whole number (c) $\mathrm{LCM}(A, B)$ in index form (d) $\mathrm{LCM}(A, B)$ as a whole number

PART B · PROBLEM-SOLVING & APPLICATIONS

Write clear mathematical statements with units

Q5. At Box Hill bus interchange, Route A departs every 15 minutes and Route B departs every 24 minutes. If both buses depart together at $7{:}00\ \mathrm{am}$, how many minutes pass before they next depart together, and at what time does this occur?

Q6. A landscape gardener in Bendigo has 84 native grasses and 126 kangaroo paw plants. She wants to plant them into the largest possible number of identical garden beds with no plants left over. Find (a) the greatest number of beds and (b) how many of each plant go into every bed.

Q7. Two interlocking spur gears have 32 teeth and 40 teeth respectively. After how many minimum rotations of the 32-tooth gear will the exact same pair of marked teeth meet again?

Q8. A rectangular sheet of plywood measuring $168\ \mathrm{cm}$ by $120\ \mathrm{cm}$ is to be sawn into identical, largest-possible square boards with no waste. Find the side length of each square and the total number of squares produced.

PART C · REASONING & CHALLENGE

Justify your conclusions using algebraic or geometric reasoning

Q9. Using the prime factorisation $252 = 2^2 \times 3^2 \times 7$, find the smallest positive integer k such that $252k$ is a perfect square, and find $\sqrt{252k}$.

Q10. Two positive integers have $\mathrm{HCF} = 14$ and $\mathrm{LCM} = 210$. Given that one of the integers is 42, use the identity $a \times b = \mathrm{HCF}(a, b) \times \mathrm{LCM}(a, b)$ to find the other integer.